

Head Trauma Evaluation Via Telemedicine

Pediatric and Adult Patients

Telemedicine is increasingly the first point of contact for patients with head trauma. While most minor head injuries do not result in clinically significant intracranial injury, clinicians must quickly identify patients who require emergency evaluation while safely reassuring those who do not.

The following cases and questions highlight a practical, evidence-based approach to telemedicine head trauma assessment using pediatric and adult decision rules.

Pediatric Clinical Case Quiz



A 10-year-old boy is evaluated via video visit after falling off his bicycle. He was not wearing a helmet but cried immediately after the fall. There was no loss of consciousness. He now reports mild headaches and nausea, but no vomiting, seizure, or abnormal behavior. His parents ask whether he needs to go to the emergency department.

What is the most appropriate next step?

a. Send the patient to the emergency department for imaging

b. Order an urgent head CT scan

c. Observe at home with clear return precautions

d. Prescribe anti-nausea medication and allow return to sports

The answer is c.

This child has normal mental status, no vomiting, and a low-risk mechanism of injury, placing him in the low-risk category under the PECARN head injury rule. Observation at home with clear return precautions is preferred over CT imaging.

Clinical Pearl

Immediate crying and normal behavior after injury are reassuring signs in pediatric head trauma.

Adult Clinical Case Quiz

A 72-year-old man presents via telemedicine after tripping on a curb and striking his head. He did not lose consciousness but reports a worsening headache and feeling “foggy.” He takes apixaban for atrial fibrillation. His spouse notes that his responses seem slower than usual.

What is the most appropriate next step?

a. Send to the emergency department for urgent evaluation and imaging

b. Reassure and observe at home

c. Treat as a concussion and arrange routine follow-up

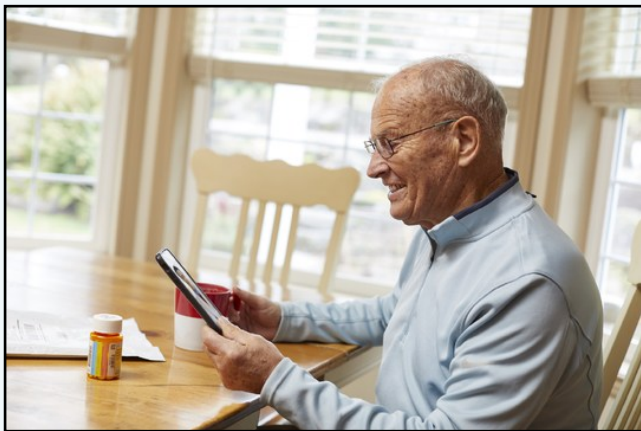
d. Schedule an outpatient head CT

The answer is a.

This patient has multiple high-risk features, including advanced age, anticoagulation use, worsening headache, and altered mental status.

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In older adults on anticoagulation, even minor head trauma warrants a low threshold for emergency evaluation.



Telemedicine Triage Quiz

Telemedicine is effective for identifying patients who do not require emergency care, but recognizing red flags is critical.

Which finding during a telemedicine visit most strongly indicates the need for immediate emergency department referral?

a. Mild headache that improves with rest

b. Isolated scalp abrasion

c. Parent concern but normal mental status

d. Abnormal mental status or confusion

The answer is d.

Any abnormal mental status following head trauma is a red flag for intracranial injury and requires urgent in-person evaluation.

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Mental status change outweighs all other symptoms in head trauma risk stratification.

Imaging Decisions in Telemedicine

Imaging for Children

In children, imaging decisions are guided by the PECARN rule. Altered mental status, signs of skull fracture, or seizure after injury require emergency department evaluation. Children with normal mental status, no vomiting, and low-risk mechanisms can often be safely observed at home.



Imaging for Adults

In adults, the Canadian CT Head Rule identifies patients at increased risk for intracranial injury. Age 65 or older, vomiting, confusion, dangerous mechanisms, skull fracture signs, and anticoagulation use increase risk substantially.

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Telemedicine is best used to identify who does not need the emergency department—red flags should always prompt escalation.

Telemedicine Head Trauma Exam: Quick Clinical Check

During a video visit for head trauma, focus first on mental status, speech, and behavior. Confirm the patient is alert, oriented, and responding appropriately. Listen for slurred speech, word-finding difficulty, or slowed responses. Observe facial symmetry, eye movements, and gross coordination. If safe, assess gait and balance. Inspect the scalp and face for visible bruise or swelling, and in children, rely heavily on caregiver reports of abnormal behavior.

Any abnormal mental status, speech change, focal neurologic deficit, worsening symptoms, anticoagulant use, or inability to ensure safe observation at home should prompt referral to the emergency department.

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When exam reliability is limited, provider uncertainty alone is a valid reason to escalate care. When caregivers report “something isn’t right,” take that concern seriously—even if objective findings are limited.

Concussion Recognition and Management

Concussion is a mild traumatic brain injury and is diagnosed clinically. Neuroimaging is often normal and should not be used to confirm or exclude concussion. Education and reassurance play a key role in recovery.

Which statement regarding concussion management is most accurate?

a. Head CT is required to diagnose concussion

b. Most concussions are diagnosed clinically, and imaging is often normal

c. Complete physical and cognitive rest is required until symptoms fully resolve

d. Patients may return to sports the same day if symptoms improve

The answer is b.

Initial management includes relative physical and cognitive rest for 24 to 48 hours, followed by gradual return to normal activities.

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Prolonged strict rest can delay recovery—early, guided activity is beneficial once symptoms improve.



Follow-Up and Safety Netting

Follow-up within 24 to 72 hours allows clinicians to reassess symptoms and detect delayed complications. Clear documentation of counseling, return precautions, and follow-up plans are essential in telemedicine care.

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Most medicolegal risk in telemedicine head trauma comes from inadequate documentation of return precautions.

Key Takeaways

Telemedicine is an effective triage tool for head trauma when used thoughtfully. The PECARN rule in children and the Canadian CT Head Rule in adults provides reliable guidance. Early recognition of red flags, avoidance of unnecessary CT scans, and clear patient education are central to safe care.

Clinical Pearl

When telemedicine creates uncertainty, the safest decision is in-person evaluation.